THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — SCIENCE · MEDIUM LEVEL FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The sex of a baby is determined at the time of 1

(a) implantation **(b)** puberty **(c)** birth **(d)** fertilisation

2. If n = sin i / sin r for glass is 1.5 and sin i = 0.75, then sin r is 1

(a) 1.125 **(b)** 0.25 **(c)** 0.5 **(d)** 0.75

3. A beam of white light is passed through a prism and a screen shows 7 colours. Replacing white by red light gives 1

(a) 7 colours **(b)** no light

(c) a single red patch **(d)** white light

4. Three resistors of 6 ohm each are connected in parallel. The equivalent resistance is 1

(a) 6 ohm **(b)** 18 ohm **(c)** 2 ohm **(d)** 3 ohm

5. A straight conductor carrying current from east to west is placed in a magnetic field pointing down. The force is directed 1

(a) to the south **(b)** to the north **(c)** downward **(d)** upward

6. In a food chain, DDT concentration is highest in 1

(a) zooplankton **(b)** small fish **(c)** phytoplankton **(d)** eagle

7. The colour of copper sulphate solution fades when an iron nail is dipped in it because: 1

(a) copper sulphate evaporates **(b)** iron turns blue

(c) $FeSO_4$ (pale green) is formed **(d)** water is formed

8. A student tests an unknown solution with litmus and gets no change. The solution is most likely: 1

(a) acidic **(b)** strongly acidic **(c)** neutral **(d)** basic

9. Iron articles are painted to prevent rusting because paint: 1

(a) keeps air and water away **(b)** makes iron reactive

(c) makes iron heavy **(d)** reacts with iron

10. Which of these is NOT a property of carbon? 1

(a) tetravalency **(b)** catenation

(c) forms multiple bonds **(d)** forms ionic bonds with chlorine

11. The blood vessel carrying deoxygenated blood from the heart to the lungs is: 1

(a) pulmonary artery **(b)** pulmonary vein **(c)** aorta **(d)** vena cava

12. A person has frequent thirst and urination and sugar in urine. Which hormone is deficient? 1

(a) thyroxine **(b)** growth hormone **(c)** adrenaline **(d)** insulin

13. How many chromosomes would a human egg cell have, if the body cell has 46? 1

(a) 46 **(b)** 92 **(c)** 23 **(d)** 12

14. A trait that is controlled by a recessive allele appears in a plant when 1

(a) both alleles are recessive **(b)** no allele is present

(c) both are dominant **(d)** one allele is recessive

15. An object is placed 40 cm from a convex lens of f = 20 cm. The image distance is 1

(a) 10 cm **(b)** 60 cm **(c)** 20 cm **(d)** 40 cm

16. If the atmosphere of the earth were absent, the day sky would appear 1

(a) blue **(b)** white **(c)** dark **(d)** red

17. The power of an electric device that draws 5 A at 220 V is 1

(a) 1100 W **(b)** 44 W **(c)** 1.1 W **(d)** 225 W

18. If a magnet is moved faster into a coil, the induced current 1

(a) becomes zero **(b)** remains the same **(c)** decreases **(d)** increases

19. Segregation of waste means 1

(a) burying

(b) separating biodegradable from non-biodegradable

(c) burning

(d) dumping in sea

20. The coefficient of O_2 in the balanced equation $C_3H_8 + O_2 \longrightarrow CO_2 + H_2O$ is: 1

(a) 5 **(b)** 3 **(c)** 7 **(d)** 4

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. Why do single-celled organisms reproduce by fission while many-celled complex organisms do not? 2

22. In pea plants, tall (T) is dominant. A tall plant is crossed with a dwarf plant and 50% dwarf offspring result. Find the genotype of the tall parent. 2

23. An object is placed 30 cm from a concave mirror. The image is formed at 30 cm on the same side. Find the focal length. 2

24. A person has a near point at 40 cm. Find the focal length of the lens needed to read at 25 cm. 2

25. How many electrons flow per second through a conductor carrying a current of 0.8 A? 2

26. State why two field lines cannot intersect each other. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. An object 6 cm high is placed 20 cm in front of a concave mirror of f = 15 cm. Find the image distance, size and nature. 3

28. A person cannot see beyond 1.5 m clearly. Find the focal length, power and nature of the corrective lens. 3

29. A battery of 9 V is connected to 3 ohm, 4 ohm and 2 ohm in series. Find the current and the voltage across the 4 ohm resistor. 3

30. A bar magnet is moved towards a coil connected to a galvanometer and then away. Describe the observations and give the reason. 3

31. In a food chain, Grass, Deer, Tiger, find the energy available to the deer and to the tiger if grass has 12,000 J. State the law used. 3

OR

Explain why the thermal decomposition of limestone needs continuous heating. Write the equation and one use of the solid product. 3

32. Explain the preparation of sodium hydroxide, naming the electrodes and products. 3

OR

Explain the concentration of ore steps for a zinc sulphide ore. 3

SECTION — D (Long Answer Type) 6 × 5 = 30

Answer in about 100–150 words each.

33. A person has a near point of 1 m and a far point at infinity. 5
(a) Name the defect
(b) find the lens power to read at 25 cm
(c) state the nature of the lens.

34. A circuit has three resistors: 6 ohm and 3 ohm in parallel, in series with 4 ohm, connected to a 12 V battery. Find 5
(a) the total resistance
(b) the current from the battery
(c) the voltage across the parallel combination
(d) the current through each parallel resistor.

35. Explain the magnetic field pattern around a current-carrying straight wire and how it changes when the current is doubled and when the distance from the wire is doubled. Describe how to show the direction of the field using the right-hand thumb rule. 5

36. Construct a food chain of a pond with five organisms and calculate the energy available at each level if the producers have 100,000 J. Which level has the least energy? 5

OR

Explain thermal, electrolytic and photolytic decomposition with one equation each. State which of the three is used in photography. 5

37. Explain the preparation and setting of plaster of Paris with equations. State its uses. 5

OR

Explain how corrosion is prevented in 5
(a) iron bridges
(b) aluminium utensils
(c) ships, with reasons.

38. Describe the substitution and addition reactions and the conditions with equations; also classify the reaction of ethene with bromine. 5

OR

Compare the human digestive system of herbivores and carnivores. 5

END OF QUESTION PAPER **Total: 80 Marks**

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